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BIOST 702
Exercise 2
09/11/2026

Question 1

Consider our research question from class: What is the average height of Duke students (in inches)? Suppose we are all-knowing, and know the true mean is 67 inches (5'7''). The true standard deviation is 3.5 inches. The distribution is normal.

Simulate 100 samples with n = 100 from this distribution. The answer to this question will simply be your code.

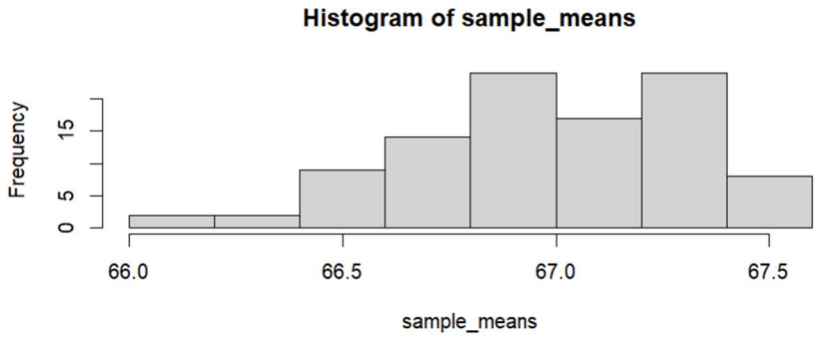
```
# True parameters
mu <- 67
sd <- 3.5
n <- 100

# Simulate 100 samples of size 100
set.seed(123)
height_samples <- sapply(
  1:100,
  function(x) rnorm(n, mean = mu, sd = sd)
)
```

Question 2

1. Calculate the sample means for each of your 100 samples. Plot these in a histogram. What does it look like?

```
##sample means
sample_means = colMeans(height_samples)
hist(sample_means)
```



: The histogram is approximately bell-shaped and symmetric, with the sample means centered around 67 inches.

2. What is the mean of your sample means? What did you expect it to be?

```
> mean(sample_means)
[1] 66.9917
```

: The mean of the sample means was approximately 66.99 inches. I expected it to be close to the true population mean of 67 inches.

3. What is the standard deviation of your sample means? What did you expect it to be?

```
> sd(sample_means)
[1] 0.3314093
```

: The standard deviation of the sample means was approximately 0.33 inches. I expected it to be close to the theoretical standard error of 0.35 inches.

4. What is the theoretical sampling distribution of the sample mean in this scenario?
: The theoretical sampling distribution of the sample mean is normal with a mean of 67 inches, a variance of 0.1225, and a standard error of 0.35 inches.

Question 3

1. Calculate the 95% Confidence Interval for each sample. The answer to this question is just your code.

```
# Standard errors
sample_se <- apply(
  height_samples,
  2,
  function(x) sd(x) / sqrt(n)
)

# Critical value
qnorm(0.975)
cv <- qt(0.975, df = n - 1)

# Confidence intervals
lower_ci <- sample_means - cv * sample_se
upper_ci <- sample_means + cv * sample_se
```

2. What percentage of your samples contain the true mean? What did you expect?

```
> #how many contain true mean?
> height_stats = data.frame(lower_ci = lower_ci,
+                           upper_ci = upper_ci) %>%
+   mutate(includes_mean = ifelse(lower_ci ≤ mu &
+                                 upper_ci ≥ mu, 1, 0))
> sum(height_stats$includes_mean)
[1] 98
> mean(height_stats$includes_mean)
[1] 0.98
>
```

98% of the confidence intervals contained the true mean of 67 inches. I expected approximately 95% because these were 95% confidence intervals. The result was slightly higher than expected due to random sampling variability.

Question 4

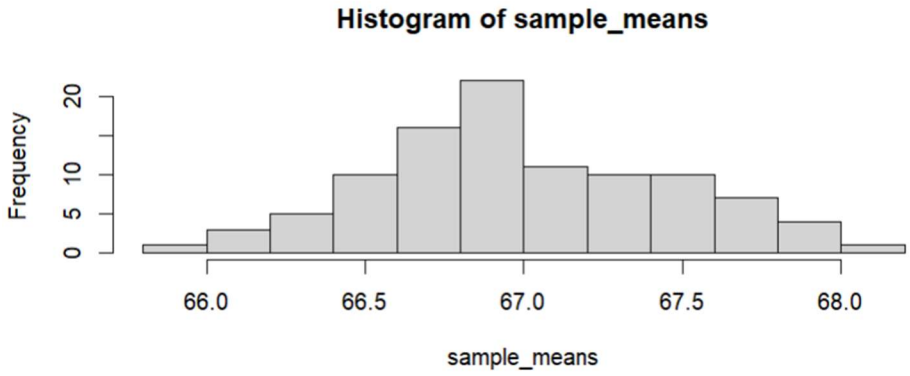
We are going to repeat everything with n = 50. Simulate 100 samples with n = 50 from the same distribution as Question 1.

1. Calculate the sample means for each of your 100 samples. Plot these in a histogram.

```r
## True parameters
mu ← 67
sd ← 3.5
n ← 50

##simulation
set.seed(123)
height_samples ← sapply(1:100, function(x) rnorm(n,mu,sd))

##sample means
sample_means = colMeans(height_samples)
hist(sample_means)
```



The histogram was approximately bell-shaped and centered around 67 inches. Compared with the histogram for n=100n=100, the sample means appeared more spread out.

2. What is the mean of your sample means? What is the standard deviation of your sample means?

```
> mean(sample_means)
[1] 66.99801
> #sd of sample means
> sd(sample_means)
[1] 0.4691038
```

The mean of the sample means was approximately 66.998 inches, and the standard deviation was approximately 0.469 inches. I expected the mean to be close to 67 inches and the standard deviation to be close to the theoretical standard error of approximately 0.495 inches. The simulated results were close to these expected values.

3. Calculate the 95% Confidence Interval for each sample. What percentage of your samples contain the true mean?

```
> sum(height_stats$includes_mean)
[1] 98
> mean(height_stats$includes_mean)
[1] 0.98
>
```

98% of the confidence intervals contained the true mean of 67 inches. I expected approximately 95% because these were 95% confidence intervals. The observed percentage was slightly higher because of random sampling variability.

4. What is the theoretical sampling distribution of the sample mean in this scenario?
The theoretical sampling distribution of the sample mean is normal with a mean of 67 inches, a variance of 0.245, and a standard error of approximately 0.495 inches.

Question 5

1. How much did the results differ from using n = 100. Why do you think this is?

   The mean of the sample means remained close to 67 inches, and the confidence interval coverage was 98% for both sample sizes. However, the standard deviation of the sample means increased from 0.33 with n=100 to 0.469 with n=50. This occurred because a smaller sample size produces a larger standard error, so the sample means vary more across samples and the confidence intervals are generally wider.

2. What do you think would happen if you increased the number of samples to 1000?

   If I increased the number of samples from 100 to 1,000, the simulated sampling distribution would become smoother and more closely match the theoretical normal distribution. The mean and standard deviation of the sample means would likely be closer to their theoretical values, and the percentage of confidence intervals containing the true mean would likely be closer to 95%. However, the theoretical standard error would not change because the sample size within each sample would remain the same.

3. What do you think would happen if you did this again with a different 100 samples? Or, if you compared with a friend, would you get the exact same results?

   If I repeated the simulation with a different set of 100 samples, I would probably obtain slightly different sample means, standard deviations, histograms, and confidence interval coverage. A friend would also not necessarily get the exact same results because the samples are generated randomly. However, we would get identical results if we used the same code and the same random seed.